

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

When the quadratic function f is graphed in the xy -plane, where $y = f(x)$, its vertex is $(-3, 6)$. One of the x -intercepts of this graph is $(-\frac{17}{4}, 0)$. What is the other x -intercept of the graph?

Answer

- A. $(-\frac{29}{4}, 0)$
- B. $(-\frac{7}{4}, 0)$
- C. $(\frac{5}{4}, 0)$
- D. $(\frac{17}{4}, 0)$

Correct Answer: B

Rationale

Choice B is correct. Since the line of symmetry for the graph of a quadratic function contains the vertex of the graph, the x -coordinate of the vertex of the graph of $y = f(x)$ is the x -coordinate of the midpoint of its two x -intercepts. The midpoint of two points with x -coordinates x_1 and x_2 has x -coordinate x_m , where $x_m = \frac{x_1+x_2}{2}$. It's given that the vertex is $(-3, 6)$ and one of the x -intercepts is $(-\frac{17}{4}, 0)$. Substituting -3 for x_m and $-\frac{17}{4}$ for x_1 in the equation $x_m = \frac{x_1+x_2}{2}$ yields $-3 = \frac{-\frac{17}{4}+x_2}{2}$. Multiplying each side of this equation by 2 yields $-6 = -\frac{17}{4} + x_2$. Adding $\frac{17}{4}$ to each side of this equation yields $-\frac{7}{4} = x_2$. Therefore, the other x -intercept is $(-\frac{7}{4}, 0)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.